



Adolescent Neurodevelopment Within the Context of Impulsivity and Substance Use

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Abstract

Purpose of Review The aim of the present review is to provide an update on recent studies examining adolescent neurodevelopment in the context of impulsivity and substance use. We provide a review of the neurodevelopmental changes in brain structure and function related to impulsivity, substance use, and their intersection.

Recent Findings When examining brain structure, smaller gray matter volume coupled with lower white matter integrity is associated with greater impulsivity across three components: trait impulsivity, choice impulsivity, and response inhibition. Altered functional connectivity in networks including the inhibitory control network and reward processing network confers risk for greater impulsivity and substance use.

Summary Across brain structure and function, there is evidence to suggest that overlapping areas involved in the rise in impulsivity during adolescence contribute to early substance use initiation and escalation. These overlapping neurodevelopmental correlates have promising implications for prevention and early intervention efforts for adolescent substance use.

Keywords Neurodevelopment · Trait impulsivity · Choice impulsivity · Response inhibition · Substance use · Adolescence

Introduction

Throughout adolescence, considered by the World Health Organization (WHO) as ages 10 through 19 [1], numerous changes in brain structure and function [2, 3], receptor expression [4], and neural connectivity occur [2, 5, 6]. Brain gray matter volume (GMV; i.e., cell bodies, dendrites) changes in an inverted U-shape: GMV increases rapidly from infancy to childhood, peaks during early adolescence, and then declines from late adolescence into adulthood [7–11]. Cortical thickness and surface area similarly decrease throughout adolescence [9, 11], together reflecting synaptic pruning and reductions in glial cells. White matter expedites

communication between various neural networks [12, 13], and white matter volume (WMV; i.e., axon bundles) and integrity increase throughout adolescence in a largely linear pattern [2]. These neural alterations are thought to result in salient cognitive, affective, and behavioral changes. Coinciding with brain changes is an evolving social context, which has increasing motivational salience among adolescents [14, 15]. Together, expected neurodevelopmental changes and navigating novel social contexts during adolescence may increase the likelihood of engaging in risky behaviors, such as substance use.

Impulsivity is often associated with risky decision-making and is frequently related to early and heavy substance use. The Diagnostic and Statistical Manual (DSM)-5 defines impulsivity in the context of a common neurodevelopmental disorder, attention-deficit hyperactivity disorder (ADHD), as “hasty actions that occur in the moment without forethought and that have high potential for harm to the individual” [16]. Beyond this broad definition, impulsivity has been further divided into various facets or components. *Trait impulsivity* reflects a personality characteristic typically measured via self-report questionnaires that can be further subdivided into

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factors including lack of premeditation, lack of perseverance, sensation seeking, and urgency [17], including positive and negative urgency [18]. *Choice impulsivity* represents a cognitive-oriented facet of impulsivity that is often conceptualized as a preference for immediate smaller rewards versus delayed larger rewards and can be measured using behavioral tasks, including delay discounting, monetary incentive delay, or Wheel of Fortune tasks [19, 20••]. *Impulsive action/response inhibition*, a motor-oriented facet of impulsivity, reflects an inability to inhibit a prepotent response, and can be assessed via self-report measures such as the Behavioral Inhibition System (BIS)/Behavioral Activation System (BAS) scale or behavioral tasks like Stop Signal or Go/No-Go [21, 22]. For review of self-report questionnaires and behavioral tasks across components of impulsivity, see [23••].

Impulsivity is elevated in various mental health disorders [24–26], including addiction [23••, 27, 28]. Components of impulsivity have been examined in relation to all phases of substance use, extending from initiation through relapse [28–30]. While numerous studies have identified impulsivity as a risk factor for early substance use initiation during adolescence [31, 32], one expanding area of research seeks to establish neurodevelopmental features underpinning impulsivity and substance use. Given the rise in transdiagnostic approaches characterizing mental health disorders, recent work has aimed to elucidate how impulsivity may serve as a transdiagnostic factor relevant to poor mental health outcomes and substance use [33•]. The aim of the present review is to provide an update on recent findings examining neurodevelopment in the context of impulsivity and substance use. Herein, we provide a review of the neurodevelopmental changes in brain structure and function related to impulsivity, substance use, and their intersection. We conclude with a discussion on how this overlap can be used as a target to inform prevention efforts and treatment interventions for adolescents.

Methods

To provide a narrative overview of neurodevelopment within the context of impulsivity and addiction, we examined the existing literature for evidence of neurobiological changes that may result in an elevated risk for impulsivity and substance use during late childhood and adolescence. We searched PubMed, Google Scholar, Scopus, and Cochrane Library. To provide a recent update on this topic, we predominantly focused on articles published within the last 5 years. However, a selection of earlier, seminal articles was included to provide foundational

knowledge and highlight important developments in our understanding of the neural basis of impulsivity and substance use during adolescence.

Neurodevelopment and Impulsivity

Brain Structure

A critical feature of neurodevelopment is the extent of neural plasticity present throughout adolescence [34]. In light of the expected brain changes occurring during adolescence, the Dual Systems Model suggests that increased risk-taking throughout adolescence is due to earlier development of reward regions of the brain, such as the striatum, yet slower development of cognitive control regions, such as the anterior cingulate cortex (ACC) and lateral prefrontal cortex (lPFC) [35]. The impact of these neurobiological changes is most salient when the socioemotional system is activated, such as in the presence of peers or situations with potential immediate rewards. In these contexts, adolescents have a greater likelihood of selecting more exciting or risky options [35]. An imbalance between the “stop”/ “top-down” inhibitory control network and “go”/ “bottom-up” reward processing network [36–38] may also underlie early substance use, as impulsivity is highly heterogeneous with a large degree of overlapping neural substrates [23••, 27, 39]. Previous studies have found only small correlations among and within the three major components of impulsivity [22]. Thus, these components likely have overlapping, yet distinct underlying brain regions [36, 40–42].

Brain Function

Alterations in functional connectivity (FC) between and within brain regions strengthen and stabilize connections to increase network efficiency throughout adolescence (for review, see [43]). Within the Dual Systems Model framework, age-related increases in connectivity between regions like the dorsolateral PFC (dlPFC) and subcortical structures like the thalamus provide further support that age-related increases in the control network reduce risk-taking behavior [44]. Specific to task-based neural activation during response inhibition, findings support distinct neural network engagement between healthy adolescents and adults, such as lower activation in frontal-parietal regions among adolescents, aside from greater activation in the ventrolateral PFC (vlPFC) [45]. Neurodevelopmental alterations in connectivity within and between brain regions may result in changes within specific facets of impulsivity and substance use.

Trait Impulsivity

Brain Structure

One of the most common measures capturing trait impulsivity among children and adolescents is the self-report UPPS-P Impulsive Behavior Scale. Items on the UPPS-P contribute to five subscale scores covering each of the trait impulsivity dimensions. Two cross-sectional reports utilized baseline imaging data from the Adolescent Brain Cognitive Development (ABCD) study to examine neural correlates of trait impulsivity using the UPPS-P [46••, 47••]; this study enrolled over 11,000 children ages 9–10. Using whole-brain approaches, positive urgency scores were negatively associated with gray matter (GM) surface area, cortical thickness, and volume in areas within the PFC (i.e., orbitofrontal cortex (OFC) and ACC), along with the amygdala, caudate, precuneus, and inferior temporal cortex [46••, 47••]. Negative urgency scores were negatively associated with surface area in the middle temporal gyrus and lateral OFC, along with cortical thickness in the left parahippocampal gyrus [47••], but were not associated with GMV [46••, 47••]. Sensation seeking was positively associated with GMV and cortical indices, including bilateral insula, frontal and occipital cortices, and precuneus, along with subcortical structures like the globus pallidus and dorsal striatum [46••, 47••]. In one ABCD study, both lack of premeditation and perseverance scores tended to have positive associations with surface area and cortical thickness in PFC regions [47••]. However, another study showed significant associations between GMV and lack of premeditation only, in a wide range of bilateral cortical and subcortical regions [46••]. Among a small sample ($n = 29$) of slightly older adolescents ages 10–14, asymmetry between the left and right medial PFC (mPFC) was associated with increased trait impulsivity, measured via parent report on the Anti-social Personality Screening Device 3 years prior [48]. Additionally, region of interest (ROI) analyses from the Pediatric Imaging, Neurocognition, and Genetics (PING) study on 328 youth ages 7–21 years found that higher total UPPS-P scores were correlated with thinner cortices in ventromedial PFC (vmPFC)/medial OFC, but not volume in the ACC or any subcortical ROIs [49].

Among adults, trait impulsivity has been associated with differences in WMI [50] but less literature has been published on adolescent samples. In one study using ABCD data, WM correlates of trait impulsivity were identified in youth. All UPPS-P subscales, aside from lack of perseverance, were associated with WMI, as measured with fractional anisotropy (FA) [47••]. Negative and positive urgency scores were negatively associated with FA

in the longitudinal fasciculus and inferior fronto-occipital fasciculus [47••]. Lack of premeditation and sensation seeking scores were positively associated with FA in the fornix, while sensation seeking scores were positively associated with FA in the anterior thalamic radiations [47••]. However, these differences in brain structure accounted for little variance in trait impulsivity scores ($< 1\%$). In sum, cross-sectional findings show that smaller and thinner GM and lower WMI in the PFC may be associated with higher trait impulsivity, specifically within the subdomains of positive and negative urgency. However, longitudinal support is needed to make causal interpretations regarding development and to understand the influence of environmental factors.

Brain Function

Trait impulsivity is correlated with FC of distinct brain regions in adolescence. Among 292 twins ages 16–28, greater impulsivity, as measured by the BIS-11, was positively associated with resting-state FC (rsFC) between the basolateral amygdala, dlPFC, and inferior frontal gyrus (IFG) [51]. These findings further support genetic overlap between neural connectivity and trait impulsivity. Consistently, evidence from clinical samples with ADHD and high-impulsivity supports impulsivity-related FC alterations among the insula and amygdala [52]. Baseline rsFC data from an ongoing longitudinal study, Neuroventure, which is following 116 adolescents aged 12–14, identified one significant ROI representing a striato-limbic network, which was positively correlated with impulsivity and negatively associated with sensation seeking, as captured via the Substance Use Risk Profile Scale (SURPS) personality traits [53]. Together, findings highlight FC correlates of trait impulsivity between the amygdala and prefrontal regions, along with the striatum and limbic regions.

Choice Impulsivity

Brain Structure

In a cross-sectional study with 400 youth (mean age = 17), delay discounting was associated with thinner cortices in brain regions involved with decision-making, such as the vmPFC, OFC, and temporal pole, and these effects were not moderated by age [54]. Similarly, two smaller studies detected negative associations between discounting for delayed rewards and GMV in areas such as the dlPFC, mPFC, and amygdala [55, 56]. Within a clinical sample, adolescents with ADHD that exhibited greater discounting of future rewards had significantly smaller GMV in areas

within the PFC, including the vmPFC, dACC, and anterior insula [57••]. Regarding WM, impulsive choice among adolescents aged 12–17 showed small-to-moderate positive correlations with WMV in prefrontal and motor cortices, but not somatosensory regions, using an ROI approach [58]. A longitudinal study among individuals ages 8–26 found age-related decreases in delay discounting (i.e., less impulsive choices across adolescence and young adulthood), and this shift was mediated by FA in the fronto-striatal tract with greater connectivity between the PFC and striatum being associated with less impulsive choices [59]. A cross-sectional study including 99 participants (mean age = 22) found no unique associations among delay discounting nor response inhibition and microarchitecture (i.e., R1 maps capturing myelin concentration) above and beyond other measures of impulsivity [60]. Discrepancies between study findings may be attributable to variations in sample size, age range, covariates, and imaging techniques. Together, these results suggest a similar pattern to trait impulsivity, such that smaller GMV and thinner cortices, and altered WMV and integrity, are associated with impulsive choice, as represented by greater discounting of future rewards during delay discounting tasks. These limited findings are mixed, and directionality may vary throughout neurodevelopment.

Brain Function

Choice impulsivity among adolescents is thought to be impacted by the social context and is associated with task-based fMRI activation in the PFC, parietal cortex, insula, and ventral striatum [61]. In a cross-sectional examination of choice impulsivity and social context ($n = 96$), younger adolescents displayed greater delay discounting by selecting more immediate rewards, and in whole-brain analyses, these choices were correlated with greater activation in a subgenual ACC/striatum cluster [61]. Older adolescents showed a preference for more delayed rewards, and these choices were correlated with greater activation in the inferior parietal lobe [61]. Regarding the social context manipulation, when adolescents were offered an immediate reward for themselves versus a delayed reward for a friend, there was increased activation in the right tempo-parietal junction (TPJ) and right precuneus [61]. However, no differences in activation were observed when the options included an immediate versus delayed reward for self or an immediate versus delayed reward for a friend. These findings align with previous research on the role of the TPJ and precuneus in social decision-making and perspective taking [62–64], and further suggest that distinct brain regions may be recruited during temporal discounting within social contexts. Among adolescents and young adults (mean age = 18), greater delay discounting was positively correlated with rsFC between the dACC and dlPFC [65]. Similarly, impulsive choice

performance was negatively associated with rsFC, particularly between the frontal pole and vmPFC, in a sample of 227 college students [41]. Among adolescents aged 13–15, the selection of smaller more immediate choices was associated with greater activation in the caudate and ventral striatum [66••]. These findings align with studies across adolescence and adulthood [67], and highlight the role of the striatum, most notably the dorsal striatum, in choice impulsivity and delayed discounting. The current literature is predominately limited by cross-sectional studies, whereas longitudinal work is necessary to map within-person changes in choice impulsivity and related brain function. Social context is another important element missing from most adolescent studies on choice impulsivity.

Impulsive Action/Response Inhibition

Brain Structure

The BIS and BAS are two motivational systems associated with GMV alterations, wherein BIS activation inhibits behavior that could result in aversive outcomes, whereas BAS activation results in approach behavior toward appetitive/rewarding cues. The BAS is thought to be enhanced during adolescence and in SUD populations [68]. The self-report BIS/BAS scale is positively correlated with behavioral tasks of response inhibition such as the Stop Signal task [69]. A cross-sectional report using BIS/BAS scale data from the ABCD study found that greater BAS scores were negatively correlated with GMV in areas including the lateral OFC, inferior frontal cortex, and precuneus [70]. Higher BIS scores, indicating greater sensitivity to aversive outcomes, were also negatively correlated with GMV in the putamen/pallidum, hypothalamus, and anterior insula [70]. A recent review focused on adolescent samples summarized evidence supporting associations between response inhibition and WMI in numerous frontal regions including the IFG and pre-supplementary motor area, and limbic system regions including the fornix and cingulum [71••]. Generally, greater FA in these areas was associated with better performance (i.e., withholding a prepotent response) on inhibition tasks [71••]. Weaknesses of the current research base include limited replicability of findings and a paucity of longitudinal studies [71••]. Within the domain of response inhibition, smaller GMV and decreased WMI appear to be associated with greater difficulty inhibiting a response among children and adolescents.

Brain Function

Neural response inhibition systems are less mature during adolescence compared to adulthood, especially regarding

effective engagement of top-down executive control systems (for review, see [72]). Behavioral differences in response inhibition between adolescents and adults, via antisaccade task performance, were associated with neural connectivity among brain regions like the dlPFC and dACC [72]. In this widely used task of inhibitory control, participants are asked to suppress a reflexive tendency to look at salient visual stimulus and quickly direct their gaze toward a stimulus in the opposite direction. In another review summarizing the neural correlates of motor response inhibition among typically developing adolescents aged 10–17, findings supported an expected pattern of age-related increases in response inhibition performance with task-based neural activation occurring in common inhibitory control network regions, such as the ventral striatum, prefrontal regions, and IFG [73•]. Generally, results show slight variations in the age-related trajectories of response inhibition performance depending on the task implemented, such that tasks likely recruit overlapping but unique neural mechanisms [73•, 74]. Data from the longitudinal cohort study IMAGEN showed that distinct patterns of neural activation were associated with both trait impulsivity and response inhibition among > 2000 healthy adolescents [75]. Specifically, among adolescents aged 14, greater task-based brain activity in the pre-sensory motor area and IFG during inhibition error processing on the Stop Signal task was significantly associated, although weakly, with lower trait impulsivity on the SURPS [75]. However, this association was not present among the same adolescents at age 19. Rather, activity in the ventral striatum during reward anticipation may be more relevant to trait impulsivity by young adulthood.

Intersection of Neurodevelopment, Impulsivity, and Substance Use

Substance use has strong and well-documented associations with trait impulsivity among youth and adults [76]. The UPPS-P urgency and lack of perseverance subscales have shown the most robust associations with substance use behavior across alcohol, cannabis, and other illicit substances, as demonstrated in a sample aged 16–26 [76]. Furthermore, longitudinal analyses among 246 adolescents aged 11–15 found that trait impulsivity in early adolescence predicted rates of alcohol use by later adolescence [77]. Shifting to impulsive choice, greater discounting of future rewards at age 14 predicted greater increases in alcohol intake over a subsequent 8-year period [78]. These associations also extend to cannabis use by late adolescence [79]. However, results across adolescence and young adulthood are mixed, as certain research findings have not supported a longitudinal association between delay discounting and future alcohol problems [80, 81].

Longitudinal studies of response inhibition have offered substantial insight into understanding how task performance, such as delayed response inhibition on Stop Signal and Go/No-Go tasks, confers risk for greater substance use vulnerability [75, 82, 83]. Yet, certain task outcomes may be less strongly related to substance use, namely, commission errors on Go/No-Go tasks (i.e., responding during “no-go” trials) were less predictive of substance use [84].

Brain Structure

Structural aberrations in brain regions associated with heightened impulsivity among adolescents are also associated with substance use. While numerous studies have noted structural consequences of early substance use [85, 86], fewer have examined structural predictors of substance use [27]. Results from whole-brain analyses of > 1500 children to young adults showed that lower GMV in the dlPFC and insula predicted alcohol use initiation [87••]. Such findings align with previous work noting that lower GMV and cortical thickness in frontolimbic regions, temporal regions, and nucleus accumbens are associated with greater risk for future substance use [88–91]. In a sample of adolescents ages 12–14 ($n = 118$), larger OFC volume, which correlated with reward sensitivity, prospectively predicted cannabis use initiation and dual alcohol and cannabis use up to 13 year later [92]. For nicotine, those who engaged in early smoking displayed smaller GMV in the anterior insular cortex compared to late-onset smokers [93]. On the contrary, across adolescents and young adults, greater GM myelination in regions including the anterior insular and subcallosal cingulate predicted a reduced risk for harmful alcohol use during a 2-year follow-up period [94••]. See Fig. 1 for GM regions that overlap between impulsivity and substance use.

Specific to WM, a longitudinal machine learning study of the IMAGEN dataset ($n = 1182$) examining predictors of alcohol misuse found that lower WMI of the corpus callosum, internal capsule, and brain stem during adolescence was predictive of alcohol misuse by young adulthood [95]. This model predicted alcohol misuse with ~ 75% accuracy. Age-related structural changes have been suggested to be synergistic in nature [96], and greater imbalance in the expected changes between GM and WM throughout adolescence, as shown by accelerated GM loss in the presence of inefficient WMI, poses a risk for greater impulsivity and early substance use initiation. While we have evidence to support the effect of structural alterations on impulsivity and substance use separately, as well as support for the effect of impulsivity on substance use, minimal research to date has examined whether impulsivity serves as a causal

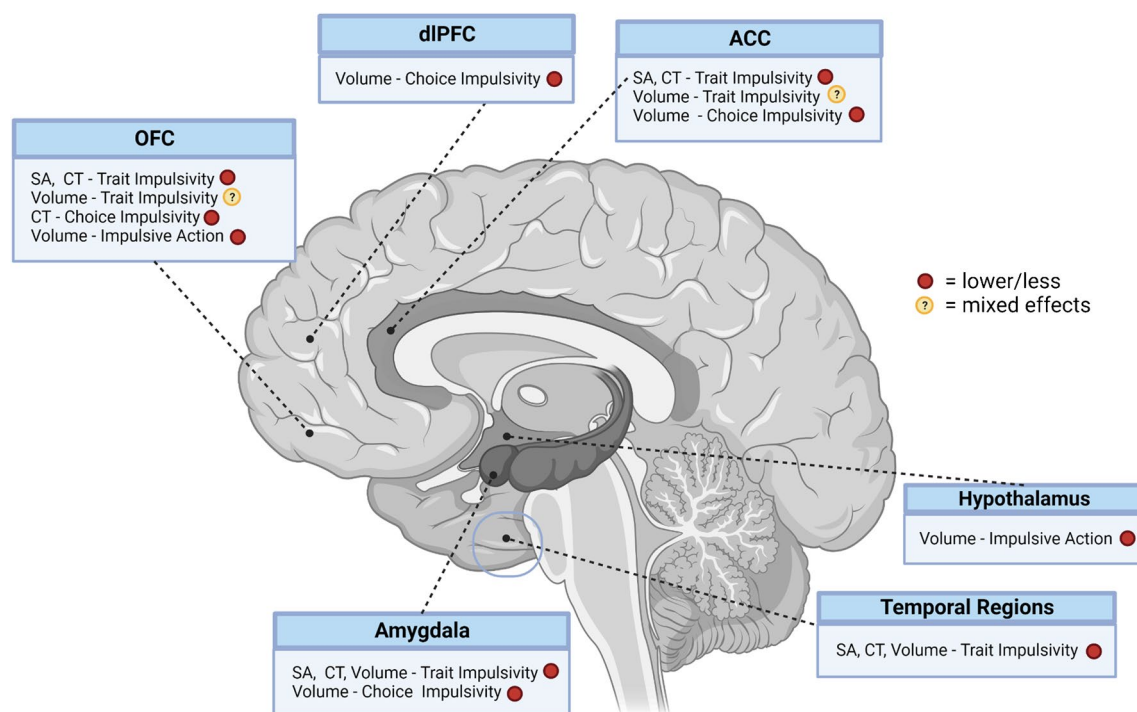


Fig. 1 Cortical structural gray matter changes associated with impulsivity during neurodevelopment that overlap with gray matter structure involved in substance use. OFC, orbitofrontal cortex; dIPFC,

dorsal lateral prefrontal cortex; ACC, anterior cingulate cortex; SA, surface area; CT, cortical thickness. The blue circle corresponds to temporal regions. Figure created in BioRender.com

mediator between brain structure and substance use in adolescence.

Brain Function

Numerous studies have demonstrated the effect of substance use during adolescence on rsFC and task-based neural activation [74, 97–99]. An increasing number of studies are beginning to examine distinct patterns of neural activation that may predate substance use [100, 101]. A recent meta-analysis examining functional neuroimaging and substance use found activation in the striatum to be one of the most robust predictors of substance use vulnerability, with at-risk adolescents displaying hyperactivation in the dorsal area of the putamen [101]. Of note, the observed differences in substance use vulnerability were primarily attributed to tasks involving a reward or motivational component, such as a monetary incentive (e.g., Wheel of Fortune task) [101], implicating neural regions involved in choice impulsivity with risk for early substance use. Altered activation in the insula and ACC similarly distinguished substance-naïve youth who later initiated substance use from those who remained abstinent [20••].

Few longitudinal studies have directly assessed the effect of impulsivity on later substance use. Among these includes a study that used magnetoencephalography (MEG) and

followed 67 alcohol-naïve adolescents [102••]. Differences in FC in the inhibitory control network distinguished adolescents who engaged in binge drinking 2 years later, compared to those engaging in light or no drinking [102••]. More specifically, adolescents who engaged in binge drinking displayed greater FC in right parietal and superior frontal areas, along with superior and temporal areas [102••]. Reductions in amygdala and OFC connectivity are linked with greater alcohol consumption during adolescence [103–105]. In a longitudinal study that enrolled 34 youth at high risk for substance use (mean age = 11), greater activation in the nucleus accumbens during expectation of monetary gain was associated with greater likelihood of substance use initiation by age 16 [106]. Within the domain of response inhibition, reduced activation in prefrontal cortical areas, and less consistently parietal and subcortical regions, during response inhibition on the Go/No-Go task has predicted alcohol and substance use progression [84]. Importantly, in the same longitudinal MEG study ($n = 67$), exploratory mediation models tested whether baseline levels of sensation seeking mediated the relationship between FC and binge drinking at follow-up [107]. Among adolescents with 2-year follow-up data, sensation seeking significantly mediated the effect of resting-state hyperconnectivity (e.g., prefrontal, medial fronto-parietal, and occipito-temporal networks) on later drinking; this model accounted for 30% of the variance.

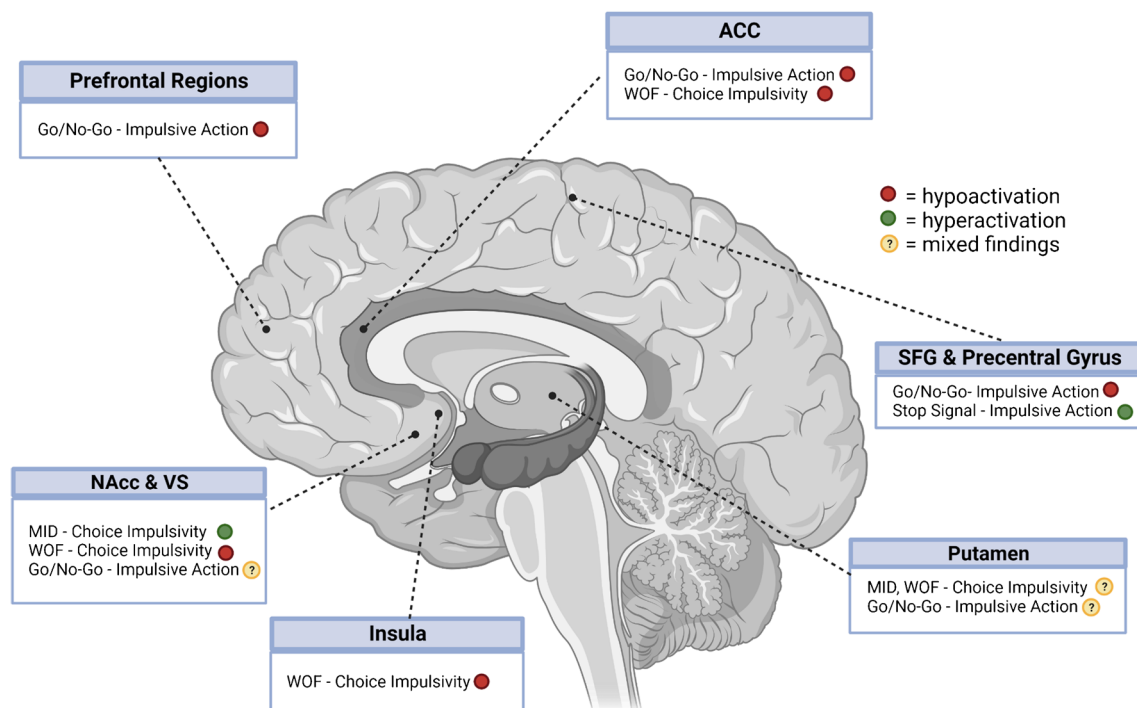


Fig. 2 Task-based functional neuroimaging regions associated with impulsivity during neurodevelopment that overlap with substance use. MID, Monetary Incentive Delay task; WOF, Wheel of Fortune

task; ACC, anterior cingulate cortex; NAcc, nucleus accumbens; VS, ventral striatum; SFG, superior frontal gyrus. Figure created in BioRender.com

Collectively, these findings highlight connectivity within and between the inhibitory control and reward processing networks that underlie facets of impulsivity and risk for early substance use initiation among adolescents. See Fig. 2 for areas of brain function that overlap between impulsivity and substance use.

Conclusions

Across brain structure and function, there is ample evidence to suggest that overlapping areas involved in the rise in impulsivity during adolescence also contribute to early substance use experimentation. Smaller GMV coupled with lower WMI is associated with greater impulsivity across the three components of impulsivity, as well as increased risk of early substance use. Reduced connectivity within and between networks involved in cognitive control and reward processing may further increase impulsivity and drive substance use experimentation. These findings both align and contradict what we would expect based on the Dual Systems Model. Across facets of impulsivity, there was evidence of earlier development of both reward processing and cognitive control regions, defined as reductions in GM (e.g., in striatum, ACC, and PFC regions), that were associated with greater impulsivity.

Functional neuroimaging findings also demonstrated a combination of hypo- and hyperactivation in regions associated with reward processing and cognitive control. These mixed findings may be due to study heterogeneity in regard to sample age and methodological approaches (e.g., task paradigms, imaging methods, statistical analyses). They also suggest that the rapidly developing reward system may continue to override a more mature “brake” system resulting in impulsive behaviors, particularly in contexts where the socioemotional system is activated. However, as there are few longitudinal studies examining these factors, we cannot confidently distinguish these effects from pre-existing neurobiological vulnerabilities.

Overlapping neurodevelopmental correlates of impulsivity and addiction have potential prevention or early intervention implications. Prevention programs including skills aimed at improving cognitive control or decision-making may serve to reduce risk for early substance experimentation and psychopathology [108–111]. Yet, mixed findings in this area suggest more research is necessary to better leverage our understanding of neurodevelopment to help delay substance use initiation. Medication or noninvasive brain stimulation methods targeting these important neural regions or treatments that improve cognitive control and functioning show promise in mitigating heavy substance use among adolescents [112, 113]. Both repetitive transcranial

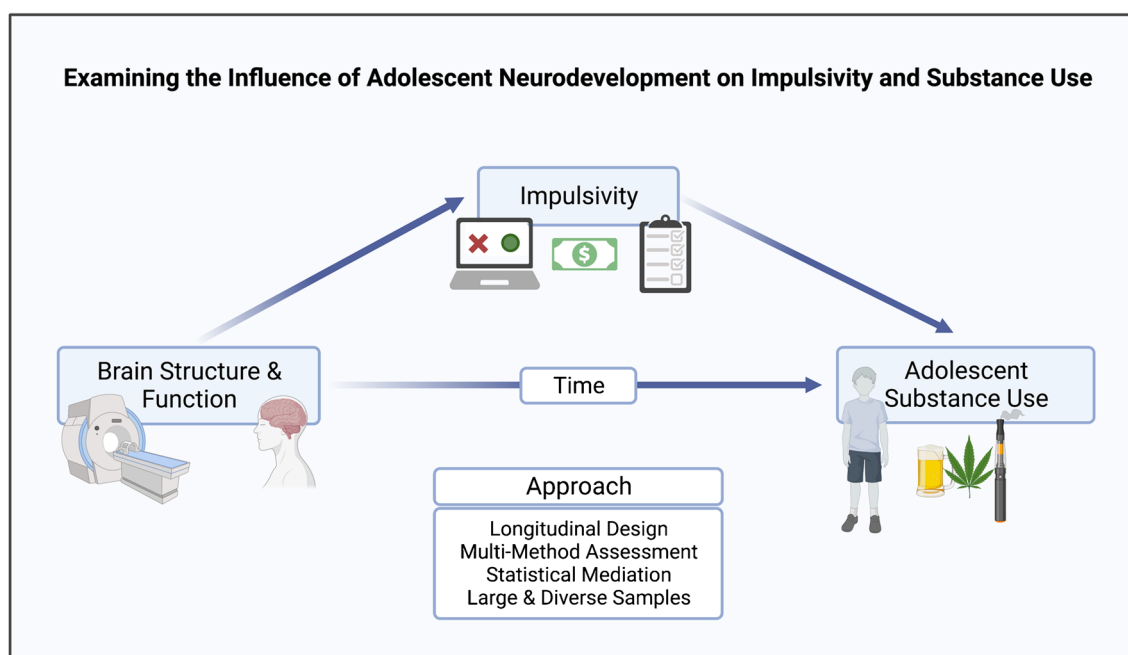


Fig. 3 Conceptual diagram of the proposed mediation pathway between neurodevelopment and adolescent substance use, including approaches to capture this pathway. Figure created in BioRender.com

magnetic stimulation (TMS) and transcranial direct current stimulation (tDCS) are two noninvasive brain stimulation methods that have been established as safe among youth [114], and may serve as potential treatment options in adolescent populations.

Notwithstanding, gaps remain in our understanding of the neurodevelopment underlying both impulsivity and addiction. As many studies have focused on GM in relation to impulsivity and substance use, future studies addressing WM structure and integrity are warranted. Published literature has highlighted challenges in defining and measuring impulsivity, specifically among adolescents and young adults [115]. Results from the current review note similar issues nearly 10 years later that can cause challenges when synthesizing results for meta-analytic work. While each component of impulsivity highlighted herein has been associated with substance use, it is unclear the relative strength of each component in predicting clinically relevant outcomes. Through measuring multiple aspects of impulsivity with multiple methods (e.g., self-report, behavioral, neuroimaging) within a longitudinal design, we may be able to better understand which components are most strongly correlated with substance use and worth targeting in the context of treatment. Implementation of longitudinal designs, with formal statistical test for mediation, is crucial to determine the extent to which alterations in brain structure and function precede adolescent substance use, and whether this is mediated by facets of impulsivity. The use of noninvasive neuroimaging approaches, such as MEG [116], and effective

connectivity approaches among child and adolescent populations may provide a more robust examination of coupling between brain regions [117], and facilitate our understanding of causal mechanisms underlying impulsivity and substance use. See Fig. 3 for conceptual diagram of a proposed mediation pathway between neurodevelopment and substance use.

Impulsivity has been established as a well-known risk factor for adolescent substance use. Recent evidence demonstrates that neural correlates of heightened impulsivity have a moderate degree of overlap with areas implicated in adolescent substance use. Most of the evidence base comes from cross-sectional studies, with relatively fewer longitudinal studies to date examining impulsivity in relation to brain structure and function over the course of childhood and adolescence. Future longitudinal, large-sample studies in this domain will help integrate the role of impulsivity as both a consequence of altered brain structure and function and a predictor and mediator of future adolescent substance use.

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Declarations

Conflict of Interest LRM reported consulting for Friends Research Institute Inc. RG, LRM, and LMS reported receiving grant support from the National Institute on Alcohol Abuse and Alcoholism (NIAAA). All other authors declare that they have no conflict of interests.

Human and Animal Rights and Informed Consent This article does not contain any studies with human or animal subjects performed by any of the authors.

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- Of major importance

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